

Doc. 313 – No Beginning:
The Thermal History on the Time Cycle

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Doc. 313 – No Beginning

What this is about

Doc. 312 ends with obligation D(ii): can the field configurations close periodically on the compact time cycle ($\tau_c = 2\pi/H_0 = 91.9$ Gyr) *while* carrying the observed thermal sequence – nucleosynthesis, recombination, structure formation? A circle has no beginning; the observations show a gradient. This document computes the question (ffgft_313_zeitzyklus_geschichte.py) and finds three results and one hard open core:

1. **Antipode finding [K|P20]:** the entire observable history fills, to 1–2 %, exactly *one half-cycle*. The last-scattering surface sits at $\theta = 0.992\pi$, the particle horizon at 1.012π – the “beginning” is the antipode of the time cycle, not an event.
2. **Smooth closure [K]:** the observed mass-drift profile reaches the antipode with slope $\sqrt{\Omega_r} \approx 0.01$ – the smoothness condition of an even periodic closure is met to 1 %, and the return half is exactly the photonically unobservable half.
3. **Entropy obstruction [K]:** exact τ_c periodicity is not dynamically generic (Poincaré discrepancy $\sim 10^{10^4}$); it must be imposed as a boundary condition. Fine-grained admissible, coarse-grained open – the sharpened core of D(ii).

Not part of the A-series; registration in Doc. 190 separately. References: Doc. 306 ($\partial T^4 = \emptyset$), 267 (degeneracy), 312 (closure scale, compact time direction, statics of the relation, torus rest frame).

Chapter A

No boundary, no $t = 0$: the starting position

FFGFT's position on singularity and big bang stands since Doc. 306: $\partial T^4 = \emptyset$ – the torus has no boundary, "what was at $t = 0$?" is wrongly posed. Doc. 312 made this concrete and gave it three supports:

Geometric: the time direction is a cycle of length 91.9 Gyr. A closed circle has no starting point – " $t = 0$ " is as meaningless on it as "the beginning of the equator".

Backwards: in the mass-drift reading (Doc. 312, statics of the relation; Wetterich), running back does not send distances to zero but masses: $\tilde{T} \cdot m = 1$ then means $\tilde{T} \rightarrow \infty$ – no density divergence, but a regime of ever slower clocks. The "singularity" is where the clocks stand still, not where the density diverges.

Forwards: no collapse into a singularity – the winding/momentum minimum (appendix of Doc. 312) has $V'' > 0$.

What Doc. 312 left open is the *history*: observations show a clear thermal sequence. How does a gradient live on a circle? That is D(ii), and here it is computed, not narrated.

Chapter B

The antipode finding [K|P20]

B.1 Where does the history sit on the cycle?

Position on the cycle: $\theta = \chi/R_H$, with χ the light path distance (the $z \rightarrow \chi$ assignment takes the measured Λ CDM map; only dimensionless ratios enter, Doc. 267). The antipode lies at $\theta = \pi$, i.e. $\chi = L^*/2 = \pi R_H = 14.09$ Gpc. Computed:

	χ [Gpc]	fraction of $L^*/2$
Recombination ($z = 1090$)	13.98	0.992
Particle horizon ($z \rightarrow \infty$)	14.27	1.012

z	θ/π	$m(z)/m_0 = 1/(1+z)$
0.5	0.140	0.667
1	0.243	0.500
3	0.465	0.250
10	0.689	0.091
100	0.917	0.0099
1090	0.992	0.00092

Finding: the entire observable history – up to $z \rightarrow \infty$ – fills one *half-cycle* to 1.2 %. The last-scattering surface sits at the antipode (99.2 %), the formal “beginning” 1.2 % behind it. The “big bang” is thereby a *place on the cycle* – the antipodal region, in the mass-drift reading the region of smallest masses and slowest clocks – and not an event before time.

B.2 Honest placement of the coincidence [S]

In Λ CDM language the same fact is the known quasi-coincidence $\eta_0 H_0/c \approx \pi$ (here 3.18 against 3.14, i.e. 1.2 %) – there an accident of the measured Ω values. In the compact reading it becomes a structural statement: *the scattering wall sits at the antipode*. For it to be more than a re-labelled coincidence, the location would have to follow forward – e.g. from the matter content binding the wall position to the half-cycle. That is not shown; status [S], noted as a derivation target.

Chapter C

The smooth closure [K]

C.1 The profile and its endpoint slope

Periodicity requires the mass drift $m(\theta)$ to close continuously on the circle. The simplest closure is an *even* profile about the antipode: minimum at $\theta = \pi$, symmetric return on the other half. It is smooth if $dm/d\theta \rightarrow 0$ at the turning point.

In the radiation era $H \simeq H_0\sqrt{\Omega_r}(1+z)^2$, hence analytically

$$\eta_0 - \chi = \frac{c}{H_0\sqrt{\Omega_r}} \frac{1}{1+z} \quad \Longrightarrow \quad \left| \frac{dm}{d\theta} \right|_{\text{end}} = \sqrt{\Omega_r} \approx 0.0096.$$

Finding: the observed profile runs into the endpoint with slope ~ 0.01 – the smoothness condition of an even periodic closure is met to 1 %. The deformation required relative to the measured history is a 1 % effect at the outermost edge.

C.2 Why this is not self-deception

The return half – where the mass drift would have to rise again – is *exactly* the half behind the scattering wall: photonically unobservable in principle. That is suspiciously convenient at first sight, but it is not a choice of this document; it is a consequence of the antipode finding: the wall *lies* at 0.992π . The closure therefore contradicts no single photon observation – and it is nevertheless not untestable (Ch. E).

The climate-zone picture from the Doc. 312 discussion thereby becomes quantitative: the gradient is the map *along* the cycle (like climate zones along a meridian circle), not an evolution *out of* a beginning – and the “hottest” point of the map is the antipode.

Chapter D

The entropy obstruction: the core of D(ii) [K]

D.1 Periodicity is not generic

The entropy budget of the observable volume is $S \sim 10^{104} k_B$ (dominated by supermassive black holes; photons: $10^{89} k_B$). Poincaré recurrence of a generic state takes $\sim e^{S/k_B} \sim 10^{10^{104}}$ dynamical times – against $\tau_c \sim 10^{18.5}$ s. The discrepancy is $\sim 10^{104}$ *orders of magnitude in the exponent*: exact τ_c periodicity cannot be expected dynamically.

Consequence: periodicity is a **boundary condition** – a selection of periodic solutions from the solution space, not an outcome of evolution. It belongs to the same category as the choice of the T^4 topology itself ([STIPULATION] in the sense of Doc. 311) and must be booked as such.

D.2 Fine- and coarse-grained, separated

Fine-grained [K]: under unitary evolution the fine-grained entropy is constant; nothing microscopic forbids an exactly periodic solution. The boundary condition is admissible.

Coarse-grained [open]: the second law is a statement about coarse-grained entropy relative to an observer. On the return half it must decrease – which is observer-relative possible (reversal of the correlation direction; no one *there* would see a violation, since the thermodynamic arrow turns with it), but unproven. **Exactly this is the sharpened core of D(ii):** not the field periodicity (which is consistent as a boundary condition, Ch. C), but the proof that a coarse-grained entropy closure is compatible with the observed structure formation. Until then: open, without embellishment.

Chapter E

Test surfaces [S]

Photons end at the wall (0.992π). Two messengers originate at or behind the antipode:

Messenger	Decoupling	Position
C ν B (neutrino background)	$z \sim 6 \times 10^9$	$\theta \approx 1.01\pi$
primordial GW	nominally $z \sim 10^{25}$	$\theta \approx 1.01\pi$

A *non-monotonic* signature there – the reversal of the mass drift beyond the antipode – would be the in-principle test surface of the closure. Practically beyond present measurement, but nameable; together with the 4.2' dipole offset (Doc. 312, torus rest frame) these are two concrete signatures of the compact reading.

Chapter F

The antipode condition as a prediction: Ω_m^* and the κ candidate

F.1 The inversion

Ch. B read the location of the scattering wall as a finding and honestly booked the coincidence $\eta_0 H_0/c \approx \pi$ as [S]. But the condition can be *inverted*: if the compact reading requires the particle horizon to close exactly at the antipode,

$$\eta_0 = \pi R_H \quad \Longleftrightarrow \quad \int_0^\infty \frac{dz}{E(z)} = \pi, \quad (\text{F.1})$$

then Ω_m is no longer a free parameter. The condition solves uniquely (flat, $\Omega_r = 9.3 \times 10^{-5}$) to:

$$\boxed{\Omega_m^* = 0.325} \quad \text{against Planck } 0.315 \pm 0.007 \Rightarrow +1.4 \sigma. \quad (\text{F.2})$$

The cosmic sector thereby obtains, for the first time, a *prediction for a Λ CDM parameter* rather than a reinterpretation – falsifiable: sharper Ω_m measurements (DESI, Euclid) decide whether 0.325 holds.

Declared inheritance: the calculation uses the Λ CDM *form* of $E(z)$ – including the $(1+z)^3$ term with full Ω_m . The prediction is therefore stated precisely: *the map must be shaped so that $\int dz/E = \pi$; in Λ CDM bookkeeping this reads $\Omega_m = 0.325$.* What carries the matter share of that bookkeeping is a separate question (Sec. F.3). Status hence [K|P20 $\times$ $E(z)$ form] – the same inheritance structure as in Doc. 309, one level deeper.

F.2 Consolidation: obligation B and the coincidence become one condition

From Ω_m^* follows $\Omega_\Lambda^* = 0.675$ and hence a structural candidate for the so far open order-one factor of the closure scale (Doc. 312, obligation B):

$$\kappa^* = 3 \Omega_\Lambda^* = 2.03 \quad \text{against measured } \kappa = 2.12 \text{ } (-4.4 \%). \quad (\text{F.3})$$

The same consolidation logic as with the 10^{123} problem ($\rightarrow$ P20): two open items – obligation B and the [S] coincidence of Ch. B – are reduced to *one* condition. The cross-check shows this is not circular: $\kappa = 2.12$ would require $\Omega_m = 0.294$, and then $\eta_0 H_0/c = 3.268 \neq \pi$. Antipode exactness and today's Planck κ stand in a genuine 4 % tension – test contact, not a fit.

F.3 Placements and one new open item

Ω_Λ **loses its substance status:** the “dark-energy density” is the FLRW translation of the closure scale; if the antipode condition holds, it is predicted, not fitted. **Hubble tension:** the compact reading takes an unambiguous side with the CMB-anchored value ($H_0 = 66.8$ from the ξ^{10} chain against Planck 67.4); the local SH0ES excess would then have to be a map effect of the calibration [S].

Dark matter: two regimes, one open. Dark *matter* has deliberately not been mentioned so far – and the gap belongs named. *Galactically* (rotation curves, RAR) FFGFT needs no particle DM: there the transition scale a_0 carries the load (Doc. 308); this document says nothing about that, correctly so. *At the background level*, however, the $z \rightarrow \chi$ map silently inherits the CDM share: in Λ CDM bookkeeping, $\Omega_m^* = 0.325$ consists of baryons (≈ 0.049) plus cold dark matter (≈ 0.276). What carries this share in the compact reading – actual matter, or a property of the mass-drift profile $m(\theta)$ that books as matter in FLRW translation – is unresolved and is booked as a **second new open item**. It is independent of the galactic question and must not be conflated with it.

New open item (R61 discipline): what sets the monopole temperature $T_0 = 2.725$ K? Dimensionlessly, $k_B T_0 / (m_e c^2) = 4.6 \times 10^{-10}$ – *not* an integer power of ξ ($\log_\xi = 2.41$). By the R61 standard no numerology is attempted here: the antipode temperature (equivalently the photon-baryon entropy ratio) is so far unbound and is booked as an open question of the cosmic sector.

Chapter G

Status overview

Statement	Content	Status
Antipode finding	history = one half-cycle to 1.2 %; scattering wall at 0.992π	[K P20]
Coincidence placement	$\eta_0 H_0 / c \approx \pi$: structural statement only after forward derivation	[S]
Smooth closure	endpoint slope $\sqrt{\Omega_r} \approx 0.01$; even closure met to 1 %	[K]
Return half	= photonically unobservable half; no conflict with photon data	[K]
Periodicity	boundary condition (Poincaré discrepancy 10^{104} dex in the exponent), not dynamics	[K]
Entropy	fine-grained admissible; coarse-grained closure open – sharpened core of D(ii)	open
Test surfaces	CvB/GW from the antipode; non-monotonicity as signature	[S]
Ω_m^*	antipode condition inverted: 0.325 (+1.4 σ); $E(z)$ form inherited	[K P20×E]
κ^*	= $3\Omega_\Lambda^* = 2.03$ (−4.4 %): candidate for obligation B; B + coincidence = one condition	[S]
DM (background)	$(1+z)^3$ share of the map: carrier unresolved (galactic: a_0 , Doc. 308, separate)	open
T_0	$k_B T_0 / (m_e c^2)$ not a ξ power: antipode temperature unbound	open

Balance: after this document, “no big bang” means precisely: the “beginning” is the antipode of the time cycle – a place, not an event; the observed history is one half of a map closable smoothly to 1 %; and the remaining burden lies not with the fields but with coarse-grained entropy. D(ii) is thereby not settled but reduced to its hard core.

Registration: entry into Doc. 190 will be done separately.